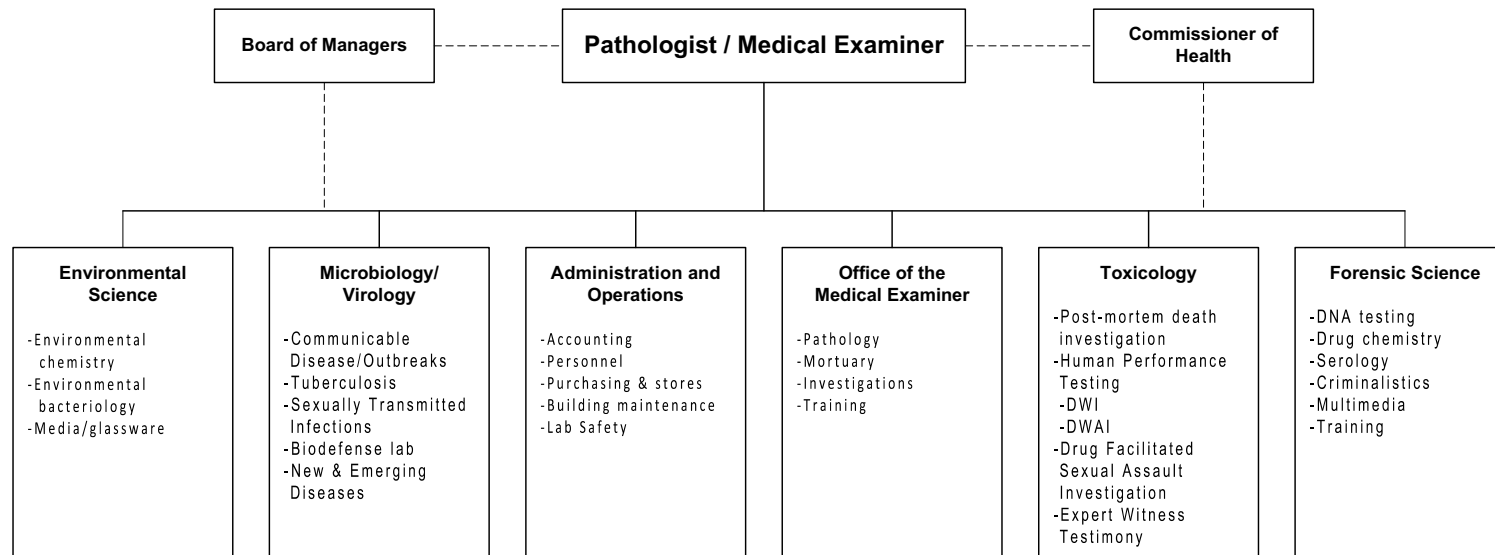


Labs and Research (31)



The Pathologist-Medical Examiner is appointed by the County Executive with the approval of the Laboratory Board of Managers and subject to confirmation by the Board of Legislators.

Labs and Research (31)

Mission Statement

The mission of the Department of Labs & Research is to conduct laboratory analysis, death investigation and professional research, in order to protect the public health and safety of Westchester County residents and visitors.

	2024	2025
POSITIONS		
Operating	105	105
Grants	4	4
	109	109
OPERATING BUDGET EXPENDITURES		
Personal Services	9,786,666	10,300,175
Equipment	200,418	100,783
Material & Supplies	1,418,728	1,363,501
Expenses	1,307,248	1,283,731
Interdepartmental Charges	4,840,736	4,535,778
TOTAL EXPENDITURES	17,553,996	17,583,968
OPERATING BUDGET REVENUES		
Interdepartmental	5,000	6,000
Departmental	2,609,708	2,662,432
State and Federal Aid	2,467,697	2,517,050
TOTAL REVENUES	5,082,405	5,185,482
DEPARTMENT TAX LEVY	12,471,391	12,398,486
Estimated Share of Miscellaneous Budget Items:		
Retirement Systems	2,352,336	2,550,892
Health Insurance and Benefits	3,701,308	3,846,470
DEPARTMENT TOTAL	18,525,035	18,795,848

The Appropriated 2024 figures in the supporting detail include all budget amendments through August 31, 2024 and Appropriations Assigned for Purchases on Order ("roll forwards") of \$146,884.

Labs and Research (31)

The Department of Laboratories and Research is New York State government's agent for the implementation and management of New York State's Program for Public Health. In order to be eligible for full State aid, counties are required to maintain a maintenance of effort for public health programs and services. The Department is required to operate under the following New York State Laws and State Codes, Rules and Regulations:

Programs and Services

Medical Examiner

Public Health Laboratories

Microbiology/Virology/Environmental Health

State Law or Regulations

NYS Consolidated Laws, Article 8 Section 400 & Section 670.

NYS Consolidated Laws, (Public Health) Section 520;

Section 2200 & 2300.

Labs and Research (31)

PROGRAM AREA AND SERVICES:

PUBLIC HEALTH & SAFETY

- ✓ Medical Examiner
 - ◆ Determine cause and manner of death in sudden, suspicious and violent cases.
 - ◆ Verify and document external causes of death with the goal to prevent them from happening
 - ◆ Extend help to grieving families in time of need.
 - ◆ Partnership with other department's divisions to provide accurate and actionable results and answers to public health or criminal justice questions.
- ✓ Division of Forensic Sciences (DFS)
 - ◆ Perform crime lab analysis such as DNA, drugs, gunshot residue, arson, trace evidence and audio/video.
 - ◆ Training of law enforcement on various aspects of evidence recognition, collection and preservation.
 - ◆ Provide expert witness testimony upon request by prosecution or defense on cases processed.
 - ◆ Provide education to county entities (Public Safety, Criminal Justice, schools) on Forensic topics
- ✓ Forensic Toxicology
 - ◆ Analyze biological samples from autopsies, DWI/DWAI and DFSA specimens for alcohol, drugs and medications.
 - ◆ Toxicology supports the Breath Alcohol Instruments and testifies in court.
- ✓ Environmental Laboratories
 - ◆ Analyze drinking water, waste water, soil, and hazardous materials for harmful chemicals and microorganisms, and for public health and environmental regulatory compliances.
 - ◆ Assisting the public for the analysis of contaminants in their drinking water.

- ✓ Microbiology/Virology/Immunology/Biodefense Laboratories

- ◆ Identify bacteria, viruses and parasites of public health concern in clinical and environmental samples.
- ◆ Maintain the Laboratory Response Network Level B Biodefense Laboratory as a national security asset to identify agents of Bioterrorism.

DEPARTMENT INITIATIVES:

NEW

- ✓ The Microbiology Lab is in the process of training three new licensed Clinical Laboratory Technologists that joined the lab in 2024. These technologists allow the lab more robust coverage and capability as testing needs in the lab return to pre-pandemic levels.
- ✓ The Forensic Division has completed the validation of two new desktop scanning electron microscopes with energy dispersive spectrometers (SEM/EDSs). These instruments increase the labs capabilities by their built-in ability to identify both traditional and non-traditional primer chemical mixtures for gunshot residue analysis (GSR).
- ✓ In accordance with the Governor's 2023 State of the State initiative to reduce gun violence crimes, the DNA database is implementing a new searchable forensics DNA index and specimen categories for use at the local and street level in New York. As a result the Forensic Division is participating in this index and will now accept swabs of guns recovered as evidence in connection with criminal possession offenses.
- ✓ Upgrading the Laboratory Information System in the Environmental Lab to a more powerful, efficient, secure and supported database.
- ✓ In accordance with EPA Lead and Copper Rule Revisions, the Environmental Division is providing analytical support to water systems with their efforts to prepare and maintain an inventory of service line materials (Lead and Copper).
- ✓ The Environmental Lab provides enhanced testing for lead in potable water systems at Westchester County Parks as proposed in the Lead Free Parks Act.

Labs and Research (31)

- ✓ The Environmental Lab is expanding accreditation to perform quantitative analysis of PFAS (Perfluoroalkyl Substances), an emerging contaminate class that is the current focus of Federal and NYS regulations.
- ✓ The Toxicology Lab will be validating a new LC-QTOF for initial screening of drugs in all cases. This will replace the existing LC-TOF. The LC-QTOF has more sensitivity than the LC-TOF which will give more confidence in screening results which will reduce turnaround times for cases.

ON-GOING

- ✓ The Forensic Division as a member of the Heroin Overdose Task Force provides analysis on overdose cases and information to the DA Intel Center.
- ✓ The Forensic Division working with NYS Investigative Genetic Genealogy (IGG) working groups supports the attempt to identify individuals such as missing persons using advance techniques.
- ✓ The Forensic Division is validating a Liquid Chromatography-Mass Spectrometer (LCMS) for marijuana and psychedelic mushroom testing, a Gas Chromatography-Mass Spectrometer with a Vacuum Ultraviolet detector for fentanyl analog analysis, and a Microvisible Spectrometer to test trace evidence such as paints and fibers.
- ✓ The Forensic Division is working with the County DA's Conviction Review Unit to determine if adjudicated cases are suitable for testing and new technologies and is also working with the County DA's Cold Case unit to assist to identify unknown victims of crimes.
- ✓ Continue to expand Forensic DNA science applications through research and new methodologies in lab automation.
- ✓ In response to the 2020 New York State Law regarding discovery and bail the Forensic division now electronically generates and sends discovery material through an electronic portal to the DA's office thus reducing time and cost of retrieving, copying and mailing thousands of documents.
- ✓ The Environmental Division performs pesticide residue analysis for NYC DOHMH Office of Vector Surveillance & Control. Pyrethrins and Pyrethroids pesticides are applied in lakes and ponds for mosquito control.
- ✓ Assist local schools in advocating for STEM disciplines by maintaining an easily accessible internship program for high school and college students by attending career days in all levels of education.
- ✓ Continue to streamline vendor instrumentation troubleshooting for clinical bacterial culture and sensitivity testing instruments via remote vendor support in the Microbiology Lab.
- ✓ Continue to utilize Matrix-Assisted Laser Desorption/Ionization- time of light (MALDI-TOF) for identification of Mycobacteria grown in culture in order to provide the Health Department with accurate and timely IDs in the Microbiology TB lab.
- ✓ The Microbiology Lab has implemented Matrix-Assisted Laser Desorption/Ionization-Time of Flight (MALDI-TOF) testing for use in identification of bacteria from culture therefore improving turnaround time and confirmatory identification.
- ✓ The Microbiology Division continues to utilize the bioMerieux Biofire instrument to provide the Health department with quick results related to gastro-intestinal illnesses.
- ✓ The Microbiology Division continues to perform real-time PCR test for the automated detection and differentiation of genomic DNA from Chlamydia trachomatis (CT) and/or Neisseria gonorrhoeae (NG) to aid in the diagnosis of chlamydial and gonorrheal disease in the urogenital tract and extragenital sites (pharynx and rectum).
- ✓ Division of Microbiological services continues to work with Westchester County Department of Health to ensure each department's needs are being met.
- ✓ Continue to support and maintain a bi-directional computer interface between microbiological services laboratory information system and County health department.
- ✓ Update current testing methodologies for agents of bioterrorism as required by revisions of National Laboratory Response Network protocols.

Labs and Research (31)

- ✓ Continue to provide COVID-19/Flu A&B/RSV PCR testing for the Medical Examiner and to respond to any clustered outbreaks that occur.
- ✓ The Environmental Lab is validating new instrumentation and method to detect Mercury in wastewater at part-per-trillion levels.
- ✓ The Toxicology Division is transitioning its drug analyses to LC/MS/MS technology moving away from GC/MS methods. LC/MS/MS methods can detect lower drug concentrations and use less chemical reagents than GC/MS methods.
- ✓ The Toxicology Division is adding “dilute and shoot” methods for the confirmation of drugs in urine for the DWI cases. This will decrease turnaround times as well as cost when compared to the traditional solid-phase column extraction methods.
- ✓ Toxicology trains ADAs on breath instruments, the effects of alcohol and other drugs in preparation for court testimony.
- ✓ Providing web based access to the Forensic LIMS that allows law enforcement agencies to track case status and receive reports electronically.
- ✓ Training the legal community in various topics in the area of forensic science.
- ✓ Moving toward a more paperless, more electronic way of storing instrument output.
- ✓ Medical Examiner's Office maintains their role as a key player, presenter, consultants and supporters of county-wide sate mandated or initiated programs including the Child Fatality Review Team and Suicide Review Team.

DEPARTMENT ACCOMPLISHMENTS:

NEW

- ✓ Maintained accreditation with National and State accrediting bodies.
- ✓ Hosted high school interns and college interns in all divisions of the department.

- ✓ Renewed state and federal grant funding to support the Forensic division, with specific focus on opioid/fentanyl testing and DNA analysis.
- ✓ The Forensic Lab underwent a full reaccreditation review to maintain accreditation with National and State accrediting bodies. The lab should be fully reaccredited by the end of 2024.
- ✓ Forensic, Environmental, Toxicology and Microbiology staff visited local county high schools to give career opportunity presentations.
- ✓ Renewal of LRN Biodefense Laboratory grant.
- ✓ The Toxicology Lab has added testing for delta-8 THC is a marijuana analog. The laboratory has identified several DWI cases with delta-8 THC.

ON-GOING

- ✓ Identified new cases of active tuberculosis, STDs and communicable diseases.
- ✓ Operation of Biodefense Lab as part of National Laboratory Response Network.
- ✓ Collaborate with CDC for identifying emerging pathogens.
- ✓ Provide surveillance testing for Chlamydia and Gonorrhea via NYS grant and other STD testing as required.
- ✓ Provide diagnostic lab testing to identify causative biological agents associated with foodborne illness communicable outbreaks and occurrences of vaccine preventable diseases.
- ✓ Maintain and optimize the use of wireless data loggers for refrigerators, freezers and incubators in Microbiological and Environmental services to provide continuous temperature monitoring and real time electronic notification of temperature excursions outside of acceptable ranges.
- ✓ Provide WCDOH with analytical support for detailed pediatric lead exposure assessments. This requires monitoring the child's environment for lead in paint, carpets, furnishings, toys, food, and removable dusts (wipes).

Labs and Research (31)

- ✓ Provide regulatory compliance monitoring to local water companies for disinfectants and disinfection byproducts, microbiological contaminants and inorganic and organic chemicals.
 - ✓ The Environmental Lab provides WCDOH with analytical support for the beach surveillance program. This program monitors for harmful bacteria after heavy rains and protects the public from swimming in contaminated bathing waters.
 - ✓ Working with Westchester Departments of Health and Emergency Services in order to provide radiological and laboratory testing assistance to support an all-hazards approach to local incident responses.
 - ✓ The Forensic Division continues operation of CODIS database for searching forensic DNA profiles locally, state-wide and nationally.
 - ✓ Remote logging in of forensic evidence by external law enforcement agencies through our new LIMS. Software enhancements to allow better communication between the Forensic Division and user agencies and the DA's office in part by providing electronic forensic reports to user agencies reducing paper, printing and mailing costs.
 - ✓ Training course provided to local police departments in various Forensic topics.
 - ✓ Use of wireless temperature and humidity monitoring system in the Forensic Division continues to give real-time information for laboratory temperatures that are out of range for a particular test.
- ✓ Grants
 - ◆ State and Federal funds to assist with Forensic Science and DNA testing.
 - ◆ State Grant to assist the operation of the Biodefense Lab for select agents, and new and emerging pathogens.
 - ◆ State Grant for surveillance testing of Chlamydia and Gonorrhea.
 - ◆ State Grants to assist with Forensic Toxicology accreditation, equipment, supplies and service contracts.
 - ◆ State Grants from Governor Traffic Safety for DWI/DWAI caseload.
 - ◆ Milk Bank revenue generated for testing of human donor milk for premature infants.
 - ◆ Department of Corrections Health Services revenue generated for STD testing.
 - ◆ TB revenue generated for TB testing for Rockland County.

REVENUES:

- ✓ Departmental Revenues - The Department develops a per test fee schedule that is updated annually.
- ✓ State Aid - Public Health costs are partially reimbursed by New York State as a percentage of costs net of revenue.

Labs and Research (31)

SERVICE INDICATORS:

SERVICE INDICATORS:

	2023 Actual	2024 Estimated	2025 Planned
Medical Examiner			
Total Deaths/Reported to ME	7,632* /3144	7,300/3,182	7,400/3,215
Death Scenes Examined	501	460	465
Autopsies Performed	365	328	328
External ME Examination	523	504	517
* CDC Wonder Database - Deaths by Underlying Cause			
Microbiology (Examples)			
TB Specimens*	4,446	4,700	4,900
STI's*	4,837	5,100	5,500
Other Communicable Diseases	826	975	1,125
*COVID-19 testing has drastically decreased			
Environmental (Water, Soil, Waste)			
Total Analytes Tested	112,749	115,610	120,000
Health Department	31,115	33,000	35,000
Environmental Facilities Dept.	71,767	74,410	76,700
Municipal Water Companies	8,367	8,200	8,300
Toxicology			
ME Submissions	550	533	550
Specimens Received from Autopsy	4,905	4,800	4,900
Law Enforcement Submissions (DWI/DWAI/DFSA)	190	175	175
Specimens Received from L.E.	380	350	350
Subpoenas for Testimony	37	20	20
Total Analyses	48,666	48,000	48,000

2023
Actual

2024
Estimated

2025
Planned

DWI submissions are down compared to pre-COVID levels for several reasons including less patrols by law enforcement due to manpower issues, an increase in defendant refusals to submit to chemical tests, and lane assistance technology that can mask the appearance of impairment.

Forensic Science

Illicit Drug Submissions / Tests*

731/28,063

862/28454

875/29,000

Arson Submissions /Tests**

12/282

4/140

5/150

Gunshot Residue Submissions/Tests

47/1892

40/2150

45/2200

DNA Analysis Submissions/ Tests

914/12,599

750/12,578

900/15,094

Trace Evidence Submissions/ Tests

170/1,838

150/1,736

165/1,910

Audio/Video Analysis Submissions

17/17

20/20

20/20

Photos taken for Forensics/ ME/Toxicology***

220/562/14

485/573/15

500/575/15

Total Analyses

44,691

45,078

48,378

*Tests can be affected due to large seizures of drugs in a single case.

**Dependent on the number of samples submitted per case.

Overall submissions in Forensics have shown a decrease, with the exception of DNA.

Labs and Research (31)

Objects of Expenditure		Expended 2023	Adopted 2024	Appropriated 2024	Projected 2024	Requested 2025	Allowed 2025
Personal Service:							
Annual-Regular:							
Office of the Medical Examiner (1000)							
Pathologist-Medical Examiner		1	1	1	1	1	1
Pathologist-Deputy Medical Examiner		1	1	1	1	1	1
Pathologist-Assoc. Medical Examiner		1	1	1	1	1	1
Sr. Medical Examiner-Investigator	XI	1	1	1	1	1	1
Medical Examiner - Investigator	X	5	5	5	5	5	5
Sr. Autopsy Asst.	IX	1	1	1	1	1	1
Histologist	VIII	1	1	1	1	1	1
Staff Asst. (Medical Examiner)	VIII	1	1	1	1	0	0
Program Specialist (ME)	X					1	1
Autopsy Asst.	VII	3	3	3	3	4	4
Office Asst.	VI	1	1	1	1	1	1
		16	16	16	16	17	17
Administrative Services (2000)							
Chief Administrator (Labs and Research)	XVIII	1	1	1	1	1	1
Program Coordinator (Env. Mgmt.)	XIII	1	1	1	1	1	1
Application Support Administrator	XII	1	1	1	1	1	1
Budget Specialist III	XII					1	1
Budget Specialist II	X	1	1	1	1	1	1
Exec. Secretary to Pathologist-ME	X	1	1	1	1	1	1
Junior Administrative Asst	VIII	1	1	1	1	0	0
Storekeeper	VII	1	1	1	1	1	1
Account Clerk	VI	1	1	1	1	1	1
		8	8	8	8	8	8
Forensic Science (3100)							
Director of Forensic Science	XVI	1	1	1	1	1	1
Sr. Forensic Scientist	XIII	4	4	4	4	4	4
Forensic Science Specialist	XII	6	6	6	6	8	8
Forensic Scientist - Multimedia	XII	1	1	1	1	1	1
Forensic Scientist	XI	11	11	11	11	9	9

Labs and Research (31)

Objects of Expenditure		Expended 2023	Adopted 2024	Appropriated 2024	Projected 2024	Requested 2025	Allowed 2025
Asst. Forensic Scientist	X	3	3	3	3	2	2
Sr Lab Tech (II - Forensic Sci)	VIII	2	2	2	2	2	2
		28	28	28	28	27	27
Toxicology (3200)							
Director of Toxicology	XV	1	1	1	1	1	1
Senior Toxicologist	XIII	3	3	3	3	3	3
Toxicologist Specialist	XII	5	5	5	5	5	5
Toxicologist	XI	1	1	1	1	1	1
Sr Lab Tech (II - Toxicology)	VIII	1	1	1	1	1	1
		11	11	11	11	11	11
Environmental Science (4100)							
Chief - Env. Laboratory Services	XVI	1	1	1	1	1	1
Office Assistant (WP)	VI	2	2	2	2	2	2
		3	3	3	3	3	3
Environmental Chemistry (4110)							
Sr. Environmental Chemist	XIII	5	5	5	5	5	5
Prog. Coord. (Env. Lab. Client Svcs.)	XIII	1	1	1	1	1	1
Environmental Chemist	XI	7	7	7	7	6	6
Asst. Environmental Chemist	X	2	2	2	2	4	4
Sr. Laboratory Technician II	VIII	4	4	4	4	3	3
Sr. Laboratory Technician I	VII	1	1	1	1	1	1
		20	20	20	20	20	20
Environmental Bacteriology (4120)							
Sr. Environmental Bacteriologist	XIII	1	1	1	1	1	1
Environmental Bacteriologist	XII	1	1	1	1	1	1